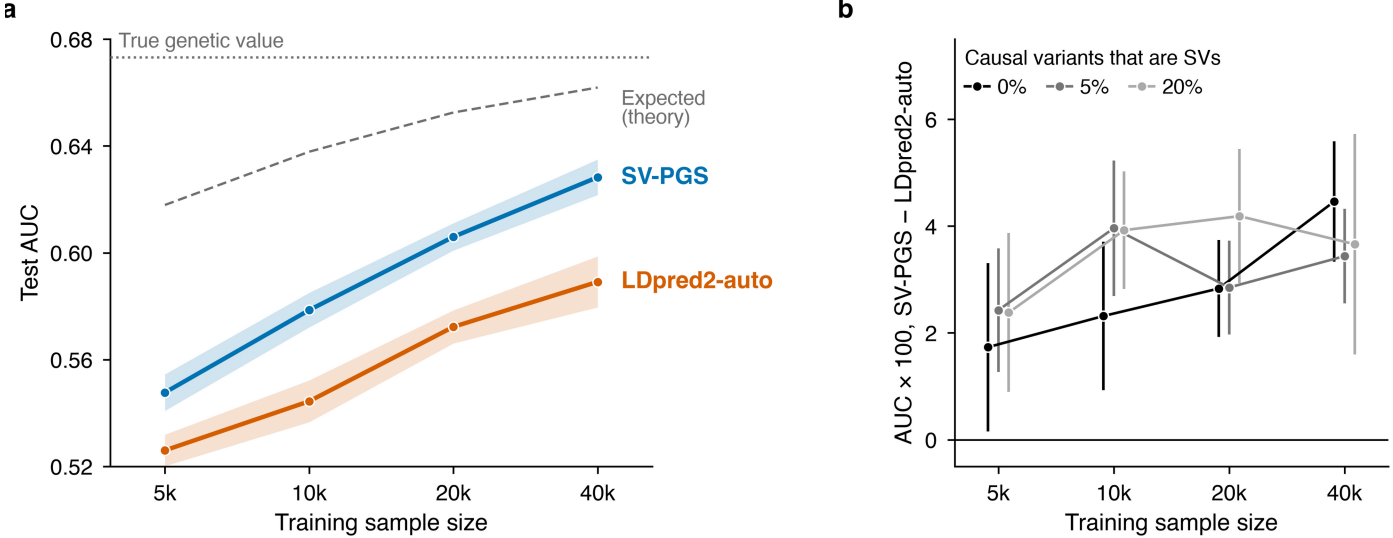


Figure 1 | SV-PGS gains accuracy faster than LDpred2-auto as the training sample grows, by the same margin whether 0%, 5% or 20% of the causal variants are SVs. A simulated binary trait on chromosome 22 (liability $h^2 = 0.1$, prevalence 10%, 200 causal variants, SV effects drawn like small-variant effects), scored in the same 10,000 held-out people by SV-PGS fitted on every SNV, indel and imputed SV and by LDpred2-auto fitted on HapMap3-density SNVs of the same training people. a, Test AUC, mean and Student-t 95% interval over the replicates of all three settings pooled (42, 37, 38, 16 replicates at 5k, 10k, 20k and 40k). Dashed: the AUC expected of a score estimated from that many people (Daetwyler et al. 2008; Wray et al. 2010); dotted: the AUC of the true genetic value. b, The paired difference in AUC between the methods on the same held-out people for each setting separately, mean and Student-t 95% interval over its replicates; the three overlap at every size, which is why a pools them. Differences are in AUC × 100 (one unit is 0.01 AUC). bench-sim v7 cohort from 1000 Genomes haplotypes, Beagle-imputed SVs; fitted on MSI.

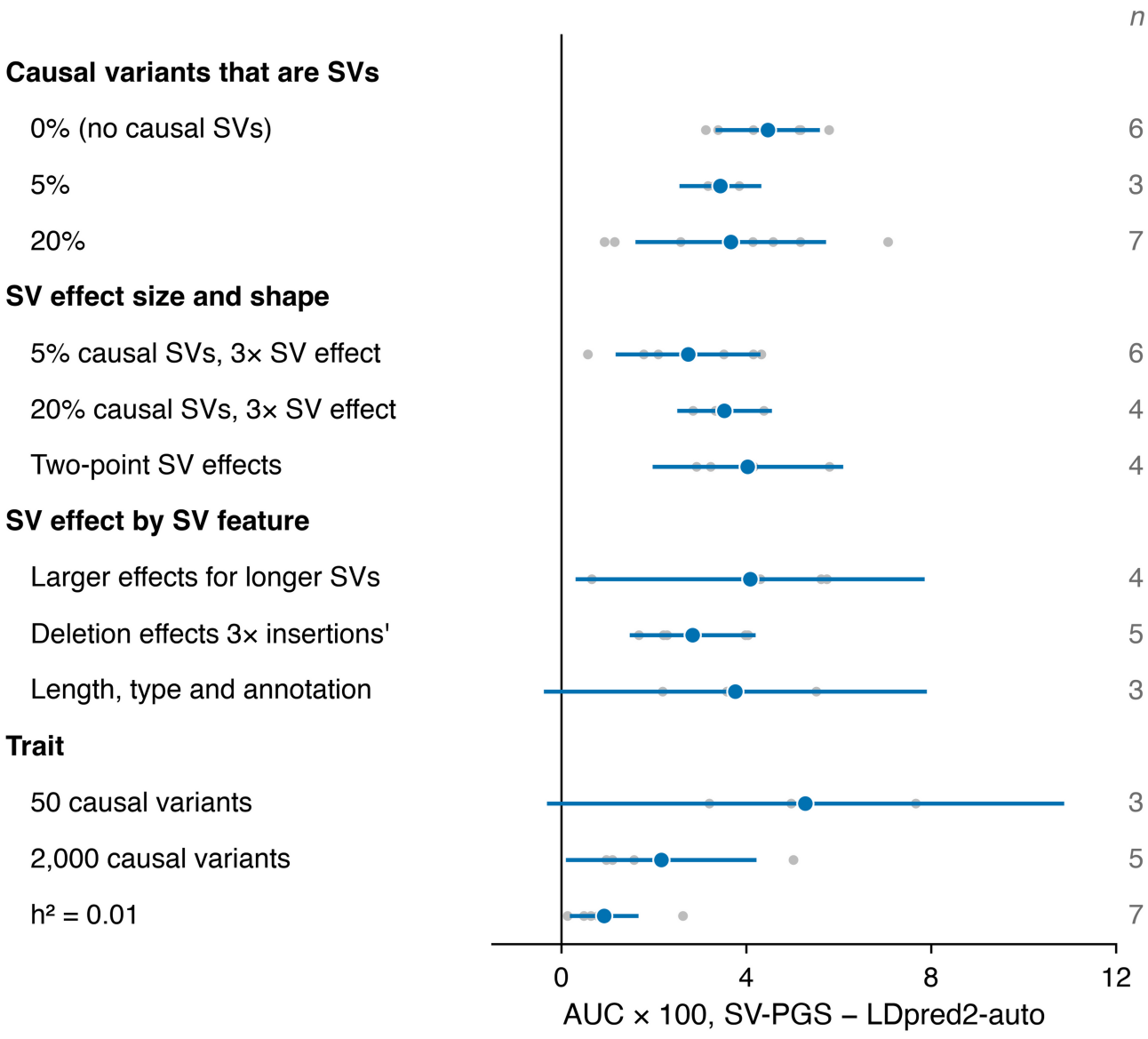


Figure 2 | SV-PGS is more accurate than LDpred2-auto across simulated genetic architectures. Paired difference in test AUC on the same 10,000 held-out people between SV-PGS (every SNV, indel and imputed SV) and LDpred2-auto (HapMap3-density SNVs), both trained on the same 40,000 people: replicates (grey), mean and Student-t 95% interval (blue); n replicates at right. SV-PGS had the higher AUC in 57 of 57 replicates. Unless named, an architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs; '3× SV effect' triples the SV effect SD, 'well-imputed' draws causal SVs only from those Beagle imputes with $r^2 \geq 0.8$ and 'common' only from SVs of frequency ≥ 5%. Differences are in AUC × 100 (one unit is 0.01 AUC). Not yet replicated three times at 40,000: heavy-tailed sv effects, only well-imputed svs, only well-imputed svs, 3× effect, only common svs, prevalence 1%.

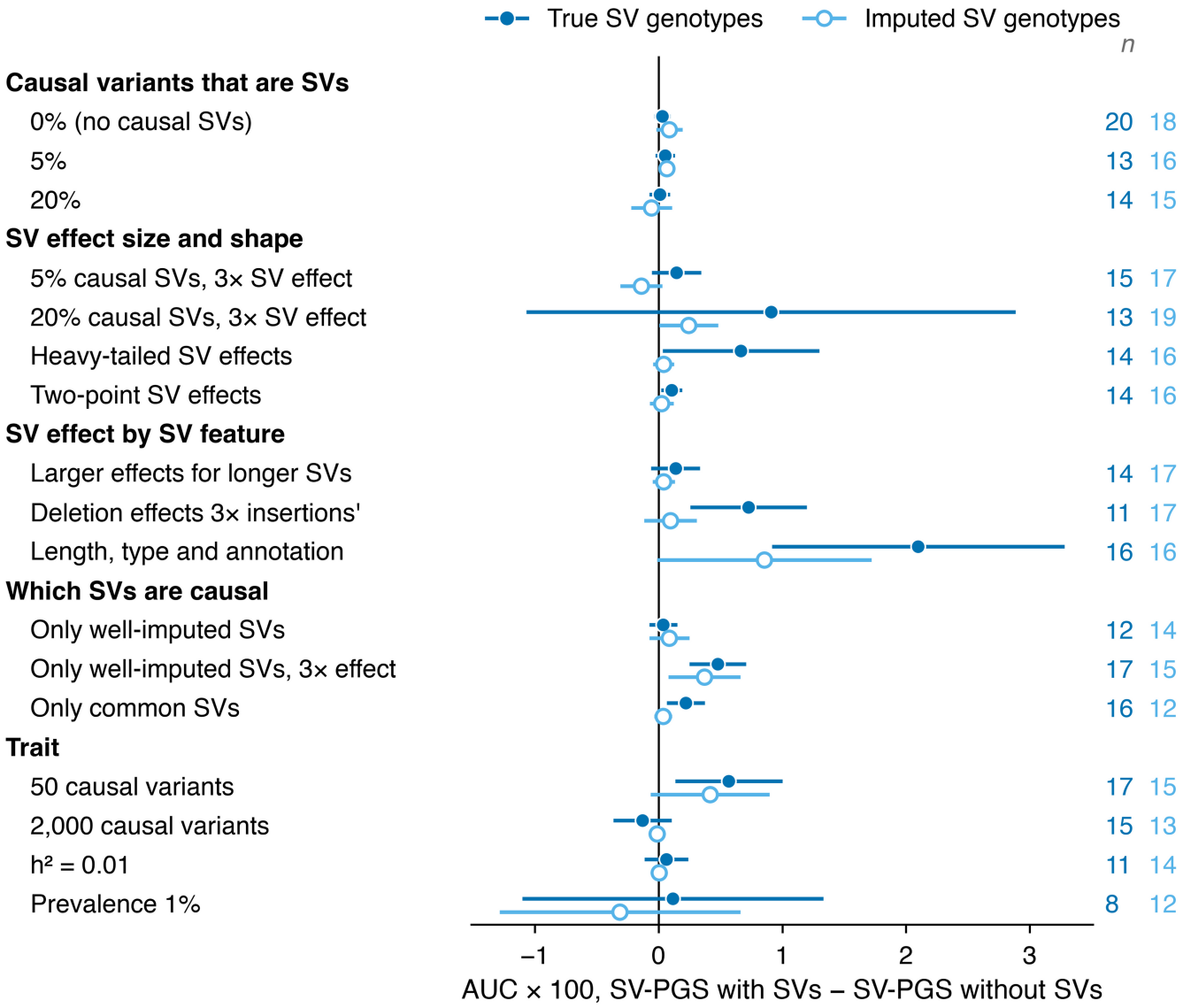


Figure 3 | SVs add accuracy when they are measured well; imputation removes most of the gain. Paired difference in test AUC on the same held-out people between SV-PGS fitted with and without SVs, when every SV is measured by its true genotype (dark) or imputed by Beagle from SNV haplotypes (light), on one 3 Mb chromosome 22 window (60,000 records). Means with Student-t 95% intervals over replicates, training sizes 5,000-40,000 pooled; n replicates at right (true, imputed). The true-genotype interval lies above zero for 7 of 17 architectures. Same simulated people, truth and fits; only the SV genotypes differ. Unless named, an architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs; '3× SV effect' triples the SV effect SD, 'well-imputed' draws causal SVs only from those Beagle imputes with $r^2 \geq 0.8$ and 'common' only from SVs of frequency ≥ 5%. Differences are in AUC × 100 (one unit is 0.01 AUC).

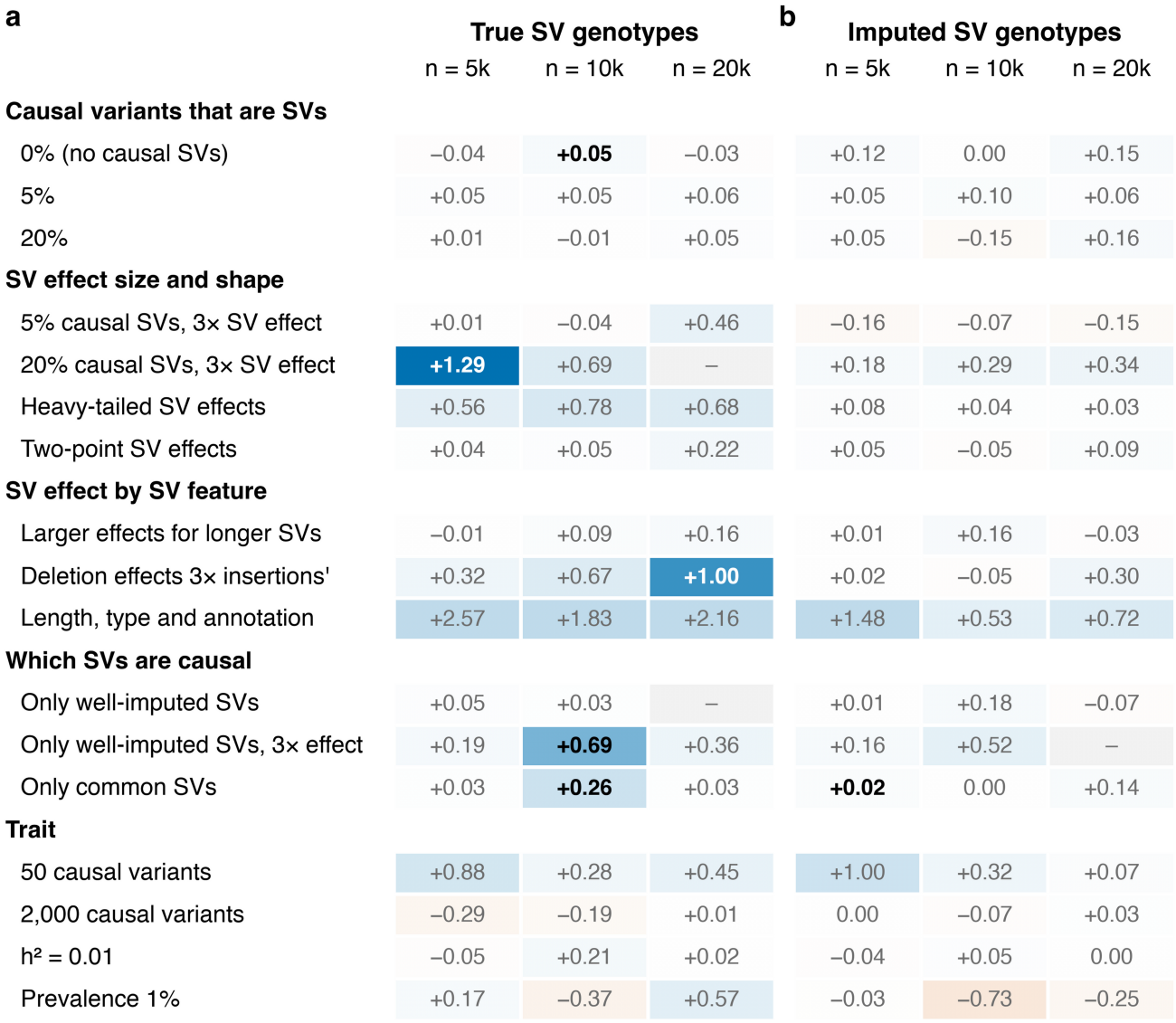


Figure 4 | The gain from adding SVs by architecture and training size. Mean paired difference in test AUC on the same held-out people between SV-PGS fitted with and without SVs on one chromosome 22 window, with SVs measured by their true genotypes (a) or imputed (b). Bold on full colour: the Student-t 95% interval over replicates excludes zero; grey on pale colour: it does not. Colour saturates at ±1.3. Dashes: fewer than two replicates. 40,000-person window fits are not yet replicated and are left out. Unless named, an architecture has liability $h^2 = 0.1$, prevalence 10%, 200 causal variants and 20% causal SVs; '3× SV effect' triples the SV effect SD, 'well-imputed' draws causal SVs only from those Beagle imputes with $r^2 \geq 0.8$ and 'common' only from SVs of frequency ≥ 5%. Differences are in AUC × 100 (one unit is 0.01 AUC).

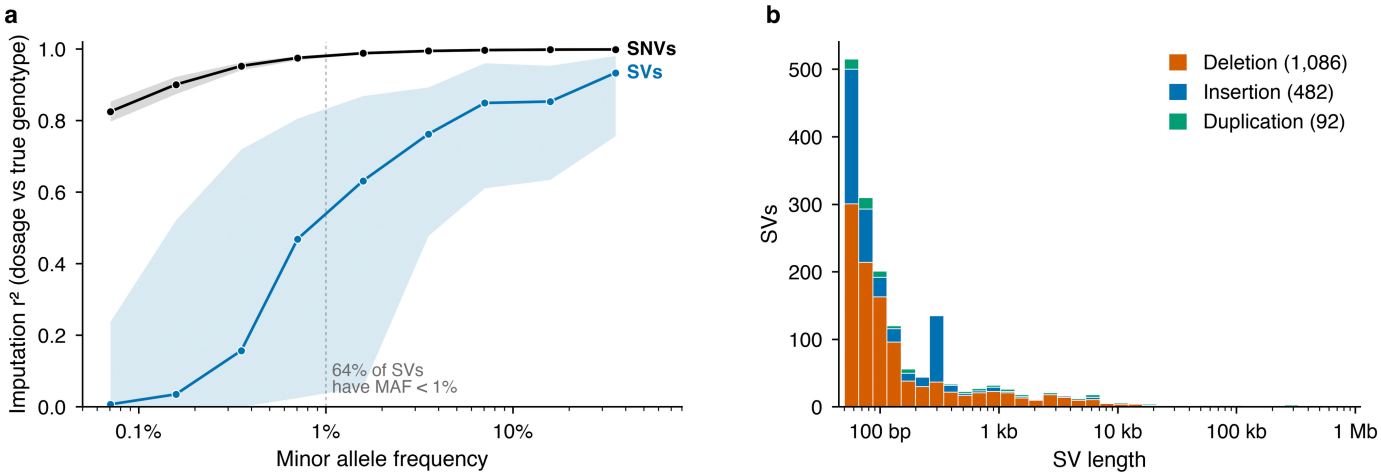


Figure 5 | The simulated SVs are mostly rare and short, and rare SVs impute poorly. a, Dosage r² between Beagle-imputed and true genotypes of 1,662 SVs and 6,000 sampled SNVs on chromosome 22 in 50,000 simulated people, by minor allele frequency: medians (points) and interquartile ranges (bands) per frequency bin. b, Lengths of the 1,662 measured SVs, stacked by class (sequence-resolved records without a type are classed by the sign of their length change). bench-sim v7 cohort from public 1000 Genomes haplotypes.

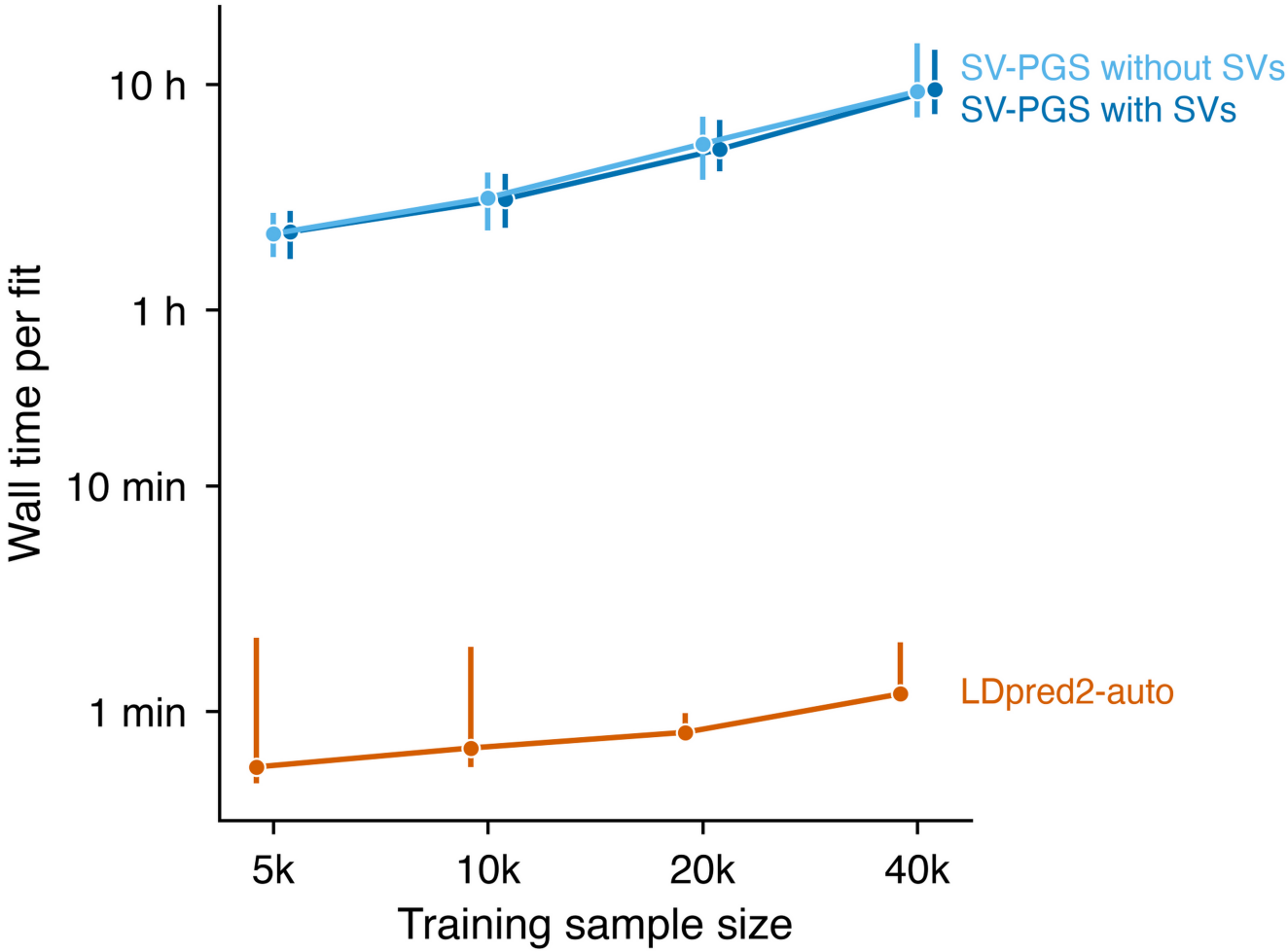


Figure 6 | Fit time. Wall time of one fit on chromosome 22 per method and training size, pooled over every architecture: median (points) and interquartile range (bars). MSI Slurm jobs, one fit per job; SV-PGS fits every variant by empirical Bayes, LDpred2-auto HapMap3-density SNVs.